Microbiology & Immunology Study Guide (Chapter 9)

Focused Review for NBDHE Preparation (Questions 1 to 50 & Clinical Cases)

How to use this guide: Study the yellow "High-Yield" callout boxes first. They contain the core microbiology and immunology facts tested directly in Questions 1 to 50. Then use the question-to-concept map, clinical vocabulary table, and rapid recall summary at the end to master your terminology.

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1. General Microbiology, Cell Structures & Staining Techniques

- **Eukaryotes vs. Prokaryotes**:
- **Eukaryotes** (Fungi, Protozoa, Helminths, Human cells): Contain a membrane-bound nucleus and specialized organelles. Protein synthesis occurs in the **ribosomes** located on the rough endoplasmic reticulum.
- **Prokaryotes** (Bacteria): Lack a nuclear membrane and membrane-bound organelles. Respiration occurs directly across the **cytoplasmic membrane**.
- **Bacterial Morphology & Staining**:
- **Morphology**: Cocci (round; clusters = *Staphylococci*, pairs = *Diplococci*, chains = *Streptococci*), Bacilli (rods), Vibrio (comma-shaped), Spirochetes (corkscrew).
- **Gram Staining**: Gram-positive bacteria stain **violet/purple** (thick peptidoglycan layer); Gram-negative bacteria stain **pink/red** (thin peptidoglycan with outer lipopolysaccharide membrane).
- **Endospore Staining**: The **Schaefer-Fulton stain** is used specifically to identify endospore-forming bacteria (*Bacillus* and *Clostridium* species).
- **Microscopy Techniques**:
- **Phase-Contrast Microscopy**: Used to examine **transparent, living cells** and observe internal cellular structures without staining.
- **Darkfield Microscopy**: Displays specimens as **bright objects against a dark background** (ideal for viewing delicate spirochetes like *Treponema pallidum*).
- **Fluorescence Microscopy**: Visualizes objects that emit light when illuminated by UV/fluorescent light.
- **Scanning Electron Microscopy (SEM)**: Provides high-resolution 3D surface images of micro-organisms.
- **Prions**: Infectious protein particles containing **no nucleic acid (DNA or RNA)**. Cause transmissible spongiform encephalopathies (e.g. Creutzfeldt-Jakob disease); highly resistant to standard sterilization.

> [!NOTE] **HIGH-YIELD FOR QUESTIONS 1, 2, 7, 8, 10, 12, 13, 14, 16, 20:** * **Ribosomes** are the eukaryotic organelles responsible for **protein synthesis**. * Bacterial cellular respiration in prokaryotes occurs across the **cytoplasmic membrane**. * Bacteria forming irregular grape-like clusters are termed **Staphylococci**. * **Phase-contrast microscopy** is used to examine **living, transparent cells**. * **Darkfield microscopy** visualizes bright specimens against a dark background. * The **Schaefer-Fulton stain** is used to detect **bacterial endospores**. * **Prions** are infectious proteins that contain **no genetic material**.

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2. Microbial Growth, Metabolism & Ecological Relationships

- **Types of Microbial Relationships**:

- **Symbiosis (Mutualism)**: Coexistence where **both species benefit** from the relationship.
- **Commensalism**: Relationship where **one organism benefits while the other is neither helped nor harmed** (e.g. normal skin flora).
- **Parasitism**: Relationship where **one organism benefits at the direct expense of the host**.
- **Predation**: One organism captures and feeds on another organism.
- **Phases of Bacterial Growth Curve**:
- **Lag Phase**: Period of intense metabolic activity and adaptation, but **no increase in cell number**.
- **Log (Exponential) Phase**: Period of rapid, exponential cell division and population growth. (Bacteria are most susceptible to antibiotics during this phase).
- **Stationary Phase**: Growth rate slows; number of new cells equals number of dying cells.
- **Decline (Death) Phase**: Nutrients depleted; **number of viable cells decreases exponentially**.
- **Energy Metabolism**:
- **Phototrophic**: Organisms that utilize light as their primary energy source.
- **Chemotrophic**: Organisms that oxidize chemical compounds for energy.
- **Hypotrophic**: Obligate intracellular parasites (e.g. viruses, chlamydiae) that rely entirely on host cell machinery.

> [!NOTE] **HIGH-YIELD FOR QUESTIONS 18, 23, 30, 31:** * **Symbiosis** is a mutually beneficial relationship between two species. * **Commensalism** is when one organism benefits and the host is unaffected. * During the **Decline (Death) phase**, the number of viable bacterial cells decreases. * **Phototrophic** organisms utilize light energy for metabolic processes.

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3. Oral Microbiology, Caries Etiology & Periodontal Biofilms

- **Dental Caries Pathogens & Demineralization pH**:
- **Smooth Surface & Pit/Fissure Caries**: Initiated primarily by **Streptococcus mutans** (and *Streptococcus sobrinus*), which produce glucans to adhere to enamel.
- **Root Surface Caries**: Primarily caused by **Actinomyces naeslundii** (and *Actinomyces viscosus*).

  Critical pH Thresholds:

  - **Enamel Demineralization**: Occurs when oral pH drops to **4.5 – 5.5**.
  - **Root Surface Demineralization**: Occurs at a higher pH threshold of **6.0 – 6.7** (cementum/dentin demineralize more readily than enamel).

- **Plaque Biofilm Composition & Successional Colonization**:
- **Initial/Early Colonizers**: Gram-positive streptococci (*S. mitis*, *S. sanguinis*, *S. oralis*) colonize the acquired pellicle. *S. salivarius* predominantly colonizes the tongue mucosa.
- **Supragingival Plaque**: Predominantly composed of **Gram-positive cocci and rods**.
- **Subgingival Plaque & Periodontitis**: Shifts toward **Gram-negative anaerobic rods and spirochetes** (*Porphyromonas gingivalis*, *Aggregatibacter actinomycetemcomitans*, *Tannerella forsythia*, *Fusobacterium nucleatum*).
- ***Porphyromonas gingivalis***: Gram-negative obligate anaerobe associated with tissue-associated plaque biofilms and severe destruction in periodontitis.

> [!NOTE] **HIGH-YIELD FOR QUESTIONS 5, 21, 22, 24, 25, 26, 27, 38, 39, 40, 41, 44, 47:** * **Streptococcus mutans** initiates enamel caries; **Actinomyces naeslundii** causes root surface caries. * Enamel demineralization occurs at **pH 4.5 – 5.5**; root surface demineralization occurs at **pH 6.0 – 6.7**. * **Gram-positive cocci** dominate **supragingival plaque**; **Gram-negative anaerobes** dominate subgingival periodontitis biofilms. * ***Porphyromonas gingivalis*** is a **Gram-negative obligate anaerobe** in tissue-associated biofilms.

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4. Viral & Bacterial Pathogens, Syphilis & Candidiasis

- **Syphilis (*Treponema pallidum*)**:
- Caused by the spirochete *Treponema pallidum*.
- **Primary Stage**: Marked by a painless, highly infectious **chancre** at the site of inoculation.
- **Secondary Stage**: Mucous patches, fever, maculopapular skin rash.
- **Tertiary Stage**: Formation of destructive **gummas**, neurosyphilis, cardiovascular syphilis.
- **Viral Pathogens**:
- **Varicella-Zoster Virus (VZV / HHV-3)**: Causes **Chickenpox** (primary infection) and **Shingles** (reactivation along dermatomes).
- **Herpes Simplex Virus (HSV-1)**: Causes primary herpetic gingivostomatitis and **recurrent herpes labialis (cold sores)**.
- **Epstein-Barr Virus (EBV / HHV-4)**: Causes Infectious Mononucleosis and **Oral Hairy Leukoplakia**.
- **Coxsackievirus A**: Causative agent of **Hand-Foot-and-Mouth disease** and **Herpangina** (painful vesicles on soft palate and tonsillar pillars).
- **Fungal Infections (*Candida albicans*)**:
- **Pseudomembranous Candidiasis (Thrush)**: Removable creamy white plaques leaving a red, bleeding surface.
- **Erythematous Candidiasis**: Smooth, flat red lesions on dorsum of tongue or palate.
- **Hyperplastic Candidiasis**: **Non-removable white plaque** typically on the hard palate or buccal mucosa.
- **Angular Cheilitis**: Redness, cracking, and soreness at the labial commissures.
- **Aphthous Ulcers (Canker Sores)**: Etiology is currently **unknown** (NOT caused by Coxsackievirus, HSV, or *S. mutans*). Presents as a round, grey-white ulcer with an erythematous halo on non-keratinized mucosa.

> [!NOTE] **HIGH-YIELD FOR QUESTIONS 6, 17, 28, 29, 35, 36, 37, 42, 43, 45, 48:** * Primary syphilis presents with a painless **chancre**. * **Varicella-Zoster Virus (VZV)** causes **Shingles** and chickenpox. * **Coxsackievirus** causes **herpangina** and hand-foot-and-mouth disease. * **Hyperplastic candidiasis** presents as a **white, non-removable plaque** on the hard palate. * **Aphthous ulcers** have an **unknown etiology** (not viral or bacterial). * **HSV-1** causes cold sores; **EBV** causes mononucleosis and hairy leukoplakia.

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5. Immunology, Immunoglobulins & Host Defense Mechanisms

- **Innate vs. Adaptive Immunity**:
- **Innate (Non-Specific) Immunity**: Present at birth, non-antigen specific, provides immediate defense, and does NOT form immunologic memory.
- **Adaptive (Acquired/Specific) Immunity**: Antigen-specific, develops after antigen exposure or vaccination, and generates **immunologic memory**.

- **Immunoglobulins (Antibodies)**:
- **IgM**: First antibody produced in response to a primary antigen challenge (**first to appear in immune response**).
- **IgG**: Most abundant serum immunoglobulin (80%); crosses the placenta to confer **passive natural immunity** to the fetus.
- **IgA**: Primary immunoglobulin in body secretions (saliva, tears, colostrum, mucus).
- **IgE**: Binds to mast cells and basophils; mediates **Type I anaphylactic/allergic reactions** (histamine release).
- **Cellular Responders**:
- **Polymorphonuclear Leukocytes (PMNs / Neutrophils)**: First immune responder cells to arrive at a site of acute inflammation/infection.
- **Infection Terminology**:
- **Bacteremia**: Presence of viable bacteria in the bloodstream.
- **Toxemia**: Presence of bacterial toxins in the bloodstream.
- **Nosocomial Infection**: Hospital-acquired or healthcare-facility acquired infection.
- **Latent Infection**: Causative agent remains inactive/dormant for a period before reactivating to produce clinical symptoms.

> [!NOTE] **HIGH-YIELD FOR QUESTIONS 3, 4, 9, 11, 15, 19, 32, 34:** * **IgM** is the first antibody to appear during a primary immune response. * **IgG** crosses the placenta conferring **passive natural immunity**. * **Neutrophils (PMNs)** are the first cells to respond to an acute infection. * **Adaptive immunity** is antigen-specific and creates **immunologic memory**. * **Toxemia** is the presence of toxins in the blood; **Bacteremia** is bacteria in the blood. * A **latent infection** remains inactive before reactivating.

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6. HIV/AIDS Manifestations, Transmission & Pharmacology

- **Transmission Routes**:
- HIV is transmitted through **sexual contact**, **exposure to infected blood/IV drug use**, and **perinatal transmission** (pregnancy, delivery, breast milk).
- **NOT Transmitted By**: Saliva, casual contact, sharing utensils, towels, or toilet seats.
- **Oral Manifestations of HIV**:
- **Oral Candidiasis**: Most common oral manifestation (occurs in ~85% of HIV patients).
- **Kaposi Sarcoma**: Purplish macules, papules, or nodules on the hard palate, gingiva, or skin (caused by HHV-8).
- **Oral Hairy Leukoplakia**: Corrugated white lesions on lateral borders of tongue (caused by EBV).
- **Linear Gingival Erythema & NUP**: Rapidly progressive periodontal destruction.
- **HIV Pharmacology**:
- Antiretroviral therapies include **Zidovudine (AZT / Retrovir)** and **Sustiva**.
- Note: Metformin is used for Type 2 Diabetes; Lorazepam for anxiety; Wellbutrin for depression.

> [!NOTE] **HIGH-YIELD FOR QUESTIONS 4, 5, 33, 34, 46, 50 & CASE STUDY:** * HIV is **NOT transmitted by sharing utensils** or through saliva. * **Oral candidiasis** is the most common oral manifestation of HIV. * **Kaposi sarcoma** presents as purplish spots on the palate, legs, or face. * **Zidovudine (AZT / Retrovir)** is an antiretroviral medication for HIV/AIDS. * **Metformin** is indicated for diabetes management.

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7. Question-to-Concept Map (Questions 1 to 50)

Question	Topic Evaluated	Key Fact to Recall
Q1	Organelle Function	**Ribosomes** are responsible for protein synthesis in eukaryotic cells.
Q2	Inflammation Signs	Bleeding is NOT one of the 5 classic cardinal signs of inflammation.
Q3	Infection Types	A **chronic infection** has a slow onset and long duration.
Q4	HIV Transmission	HIV is **NOT transmitted by sharing eating utensils**.
Q5	HIV Oral Signs	Recurrent aphthous ulcers are NOT a specific diagnostic sign of HIV.
Q6	Warts Causative Agent	**Papillomaviruses (HPV)** cause oral and skin warts.
Q7	Eukaryotes vs. Prokaryotes	**Fungi** are eukaryotic; bacteria are prokaryotic.
Q8	Microscopy Techniques	**Darkfield microscopy** shows bright objects against a dark background.
Q9	Maternal Immunity	Maternal antibodies transferred to fetus provide **passive natural immunity**.
Q10	Immune Responders	**Polymorphonuclear leukocytes (PMNs)** are the first cells to respond to infection.
Q11	Blood Toxins	**Toxemia** refers to toxins circulating in the bloodstream.
Q12	Endospore Staining	**Schaefer-Fulton stain** is used to identify spore-forming bacteria.
Q13	Bacterial Clusters	**Staphylococci** are bacteria arranged in irregular grape-like clusters.
Q14	Commensalism	**Commensalism** is where one organism benefits and the host is unharmed.
Q15	Primary Antibody	**IgM** is the first immunoglobulin to appear in an immune response.

Q16	Latent Infection	Infection where agent remains inactive before producing symptoms is **latent**.
Q17	Syphilis Primary Stage	A painless **chancre** appears during the primary stage of syphilis.
Q18	Cell Membrane Function	The **cytoplasmic membrane** regulates movement of materials into/out of cell.
Q19	Prokaryotic Respiration	Bacterial respiration occurs across the **cytoplasmic membrane**.
Q20	Shingles Causative Agent	**Varicella-Zoster Virus (VZV)** causes shingles (herpes zoster).
Q21	Enamel Caries Agent	***Streptococcus mutans*** initiates enamel caries.
Q22	Initial Infant Colonizer	*Streptococcus salivarius* is a Gram-positive initial colonizer of infant mouth.
Q23	Bacterial Growth Phase	Viable cell count decreases exponentially during the **decline/death phase**.
Q24	Bacteria in Blood	**Bacteremia** is the presence of bacteria in the bloodstream.
Q25	Hard Surface Colonizer	***Streptococcus mutans*** colonizes hard dental surfaces.
Q26	Supragingival Plaque	**Gram-positive cocci** predominate in supragingival dental plaque.
Q27	Disease Transmission	Portal of exit is required for disease transmission (not a caries prerequisite).
Q28	Aphthous Ulcer Etiology	The etiology of **aphthous ulcers** is currently unknown.
Q29	Hyperplastic Candidiasis	Presents as a **white non-removable plaque** on the hard palate.
Q30	Phototrophic Organisms	**Phototrophic** organisms utilize light energy for growth.
Q31	Symbiosis Definition	**Symbiosis** is a mutually beneficial interaction between two species.

Q32	Adaptive Immunity	**Adaptive immunity** is antigen-specific and creates immunologic memory.
Q33	HIV Transmission	HIV is **NOT transmitted through saliva**.
Q34	HIV Risk Factors	Sexual contact, blood exposure, and IV drug use are HIV risk factors.
Q35	Chickenpox Agent	**Varicella-Zoster Virus** causes chickenpox (varicella).
Q36	Fecal-Oral Virus	**Hepatitis A virus** is transmitted via the fecal-oral route.
Q37	Syphilis Etiology	***Treponema pallidum*** is the spirochete that causes syphilis.
Q38	Periodontal Pathogen	***Porphyromonas gingivalis*** is a key periopathogen (not cariogenic).
Q39	P. gingivalis Traits	***P. gingivalis*** is a Gram-negative obligate anaerobe.
Q40	Plaque Biofilm Location	***P. gingivalis*** resides in tissue-associated subgingival biofilms.
Q41	Caries Initiation	***Streptococcus mutans*** is responsible for initiating caries.
Q42	Aphthous Ulcer Appearance	Greyish-white ulcer with an **erythematous halo** on unattached mucosa.
Q43	Canker Sore Cause	Etiology of aphthous ulcers is unknown (not viral/bacterial).
Q44	Periodontitis GCF	**Gram-negative species** predominate in gingival crevicular fluid in periodontitis.
Q45	Cold Sore Agent	**Herpes Simplex Virus Type 1 (HSV-1)** causes cold sores/herpes labialis.
Q46	Kaposi Sarcoma	Presents as **purplish macules/spots** on legs, feet, face, or palate.
Q47	Root Caries Agent	***Actinomyces naeslundii*** is primarily responsible for root surface caries.

Q48	Non-Removable Candidiasis	**Hyperplastic candidiasis** presents as a white non-removable plaque.
Q49	NUP Presentation	**Necrotizing Ulcerative Periodontitis** causes gingival necrosis with gray pseudomembrane.
Q50	Diabetes Medication	**Metformin** is indicated for diabetes; Wellbutrin is for depression.

8. Clinical Vocabulary (English – Spanish)

Clinical English Term	Spanish Translation	Clinical Definition / Significance
Ribosome	Ribosoma	Organelo celular encargado de la síntesis de proteínas.
Cytoplasmic membrane	Membrana citoplasmática	Regula el paso de sustancias; sitio de respiración celular bacteriana.
Phase-contrast microscopy	Microscopía de contraste de fases	Técnica para examinar células transparentes y vivas sin tinción.
Darkfield microscopy	Microscopía de campo oscuro	Muestra espécimen brillante sobre fondo oscuro (ideal para espiroquetas).
Endospore stain	Tinción de endosporas	Tinción de Schaefer-Fulton para detectar esporas de *Bacillus* y *Clostridium*.
Staphylococci	Estafilococos	Cocos grampositivos agrupados en racimos irregulares.
Symbiosis	Simbiosis	Coexistencia mutuamente beneficiosa entre dos especies.
Commensalism	Comensalismo	Relación donde un organismo se beneficia y el huésped no se afecta.
Decline phase	Fase de declive (muerte)	Fase bacteriana donde disminuye exponencialmente el número de células viables.
Demineralization pH	pH de desmineralización	Ensamble: **4.5–5.5** para esmalte; **6.0–6.7** para superficie radicular.
Chancre	Chancro	Lesión ulcerada primaria e indolora de la sífilis (*T. pallidum*).

Shingles	Culebrilla (Herpes Zóster)	Reactivación del virus Varicela-Zóster a lo largo de dermatomos.
Herpangina	Herpangina	Infección por Coxsackievirus con vesículas dolorosas en paladar blando.
Hyperplastic candidiasis	Candidiasis hiperplásica	Placa blanca no desprendible en el paladar duro (*C. albicans*).
Aphthous ulcer	Úlceras aftosas (aftas)	Úlcera grisácea con halo eritematoso de etiología desconocida.
IgM	Inmunoglobulina M	Primer anticuerpo producido en la respuesta inmune primaria.
IgG	Inmunoglobulina G	Anticuerpo sérico más abundante; cruza la placenta (inmunidad pasiva).
PMNs / Neutrophils	Neutrófilos PMN	Primeras células de defensa en llegar al sitio de infección aguda.
Toxemia	Toxemia	Presencia de toxinas bacterianas en el torrente sanguíneo.
Kaposi sarcoma	Sarcoma de Kaposi	Lesión vascular maligna violácea oral/cutánea asociada a VIH (HHV-8).

9. Rapid Recall High-Yield Facts

- **Ribosomes** → Site of protein synthesis in eukaryotic cells.
- **Bacterial Respiration** → Occurs across the **cytoplasmic membrane** in prokaryotes.
- **Staphylococci** → Bacteria arranged in **irregular grape-like clusters**.
- **Phase-Contrast Microscopy** → Visualizes **living, transparent cells** without staining.
- **Schaefer-Fulton Stain** → Used to identify **bacterial endospores**.
- **Prions** → Infectious proteins containing **no nucleic acid (DNA or RNA)**.
- **Enamel Demineralization pH** → **4.5 – 5.5**; Root surface demineralization pH = **6.0 – 6.7**.
- ***Streptococcus mutans*** → Initiates enamel caries; ***Actinomyces naeslundii*** causes root surface caries.
- **Supragingival Plaque** → Dominated by **Gram-positive cocci**; Subgingival plaque dominated by **Gram-negative anaerobes**.
- ***Porphyromonas gingivalis*** → **Gram-negative obligate anaerobe** in tissue-associated biofilms.
- **Primary Syphilis** → Presents with a single painless **chancre**.
- **Varicella-Zoster Virus** → Causes **Chickenpox** and **Shingles**.
- **Coxsackievirus A** → Causative agent of **herpangina** and hand-foot-and-mouth disease.
- **Hyperplastic Candidiasis** → Presents as a **white non-removable plaque** on the hard palate.
- **Aphthous Ulcers** → Etiology is **unknown** (NOT viral or bacterial).

- **IgM** → **First antibody** produced in immune response; **IgG** crosses placenta.
- **PMNs (Neutrophils)** → **First immune cells** to respond to infection.
- **HIV Transmission** → Transmitted by blood, sexual contact, perinatal; **NOT by saliva or sharing utensils**.
- **Kaposi Sarcoma** → Purplish macules on palate or skin in HIV patients.
- **Metformin** → Indicated for **diabetes** management; **Zidovudine (AZT)** for HIV.